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Identification of novel non-homologous drug targets against *Acinetobacter baumannii* using subtractive genomics and comparative metabolic pathway analysis

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ABSTRACT

Lack of effective antibiotics and the development of multidrug resistance in clinical isolates of nosocomial pathogen *Acinetobacter baumannii* has necessitated the identification of novel drug targets. The study is divided into three phases, in phase I, four different sets of proteins were subjected to a chokepoint, plasmid, resistance genes, and virulence factors analysis. After phase 1 analysis we obtained two hundred twenty-two proteins which were analyzed further in the phase II for essentiality and homology. Fifty-eight proteins identified as target candidates were studied for qualitative characteristics. Among them, 32 were identified as cytoplasmic membrane, 17 as cytoplasmic, one as periplasmic, one as outer membrane, two as extracellular, and location of 5 was not known. Druggability analysis revealed that 18 proteins were druggable, and 40 were novel. Drug targets obtained in the present study can be utilized for the identification of novel antimicrobials for the treatment of infections caused by multidrug-resistant *A. baumannii*. Predicted drug targets can be evaluated for their binding affinity by molecular docking studies and thus accelerating the process of drug discovery.

1. Introduction

A. baumannii is a non-motile, oxidase negative, catalase positive, non-fastidious and non-glucose fermenting Gram negative bacterium [1]. It is a leading cause of nosocomial infections, and a common colonizer in patients admitted to healthcare facilities [1]. Antimicrobial resistance (AMR) has appeared as a significant cause of morbidity and mortality due to the increasing trend in drug resistance along with the shortage of new antibiotics [2]. A meeting on antimicrobial resistance has estimated that there will be over 10 million deaths per year by 2050 due to AMR if the current situation is not controlled [2]. Therefore, detailed knowledge of antibiotic resistance and its increasing trend will help us to understand the present scenario and also help us in making future estimates [2]. World Health Organization has designated *Acinetobacter* as a priority organism in the list of antibiotic-resistant "priority pathogens" [3]. As per the report of the CDC, *A. baumannii* infection was diagnosed in 8500 hospitalized patients and about 700 people died due to its infection in 2017 [4]. According to a study which was performed to estimate the mortality burden due to the multidrug resistant (MDR) microorganisms in India, it was found that average mortality rate due to MDR infections is 13.1% but it was two to three times higher in the case

of infections caused by MDR and extremely drug resistant (XDR) microorganisms like *E. coli*, *K. pneumoniae* and *A. baumannii* [5]. *A. baumannii* is involved in various hospital-acquired infections like bloodstream infections and ventilator-associated pneumonia [6]. The mortality rates with these infections can be as high as 35% [2]. The treatment of these infections is difficult as *A. baumannii* is MDR and can cause epidemics in the hospital setting [2]. Emerging multi-drug resistance and a lack of appropriate antibiotics has necessitated the need for new antibiotics as well as the identification of new drug targets. Target identification is regarded as the central point in the drug discovery process. In the present study, hierarchical computational approach was used for identification of drug targets in hypervirulent strain *A. baumannii* LAC-4.

The conventional method of drug discovery and development process involves expensive techniques, which are time-consuming and challenging and also identify fewer drug targets. On the other side, computational approaches have revolutionized the field of drug discovery by easier identification of drug targets utilizing OMICS data and reduce the failure rate of the drugs in clinical trials [7]. The development of potential drug targets was further facilitated by the availability of the sequenced genomes of humans and pathogens in public databases.

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Current drug target identification approaches are based on non-homologous enzymes, chokepoint enzymes, pathogen-specific essential genes, genes associated with resistance, and virulence [7]. Computational techniques have been used previously to find potential drug targets for various life-threatening pathogens, including *Mycobacterium tuberculosis* [8], *Streptococcus pneumoniae* [9] and *Helicobacter pylori* [10]. Drug target identification approaches consider essentiality and specificity as the basis for the selection of the protein as a target. A protein, which was selected as the target must be pathogen-specific so that it can prevent the undesirable interactions of drugs with the host and it must be vital for growth and survival of microorganism. Any interference with this type of protein can prove to be fatal for the microorganism one example of such study is the identification of the drug targets in the *Mycobacterium abscessus* using *in silico* approach [11]. Proteins involved in the peptidoglycan synthesis are very useful targets for the class of antibiotics like penicillin and peptidoglycan synthesis is essential for the survival of the microorganisms [12]. Based on essentiality, specificity, role in pathogenesis and development of resistance, present study was carried out for the identification of drug targets by using *in silico* hierarchical approach against *A. baumannii*.

2. Materials and methods

The present study was performed for the identification of drug targets in *A. baumannii* using the hierarchical computational approach and is divided into three phases. Flowchart (Fig. 1) demonstrates the algorithm utilized for the identification of the drug targets.

2.1. Phase 1

2.1.1. Metabolic pathway analysis

Comparative analysis of metabolic pathways between *A. baumannii* LAC-4 and *H. sapiens* was carried out using KEGG database [13]. Proteins from different metabolic pathways were subjected to BLASTp search against non-redundant database [14]. Proteins that showed no similarity were defined as non-homologous proteins and selected for further analysis. We adopted a rigorous measurement of no similarity for non-homologous proteins [15].

2.1.2. Chokepoint analysis

"Chokepoint responses" are the reactions which exclusively catalyze a particular substrate or preferentially deliver a particular product. Choke point analysis includes all reactions that raise several pathways while excludes those reactions catalyzed by more than one enzyme and

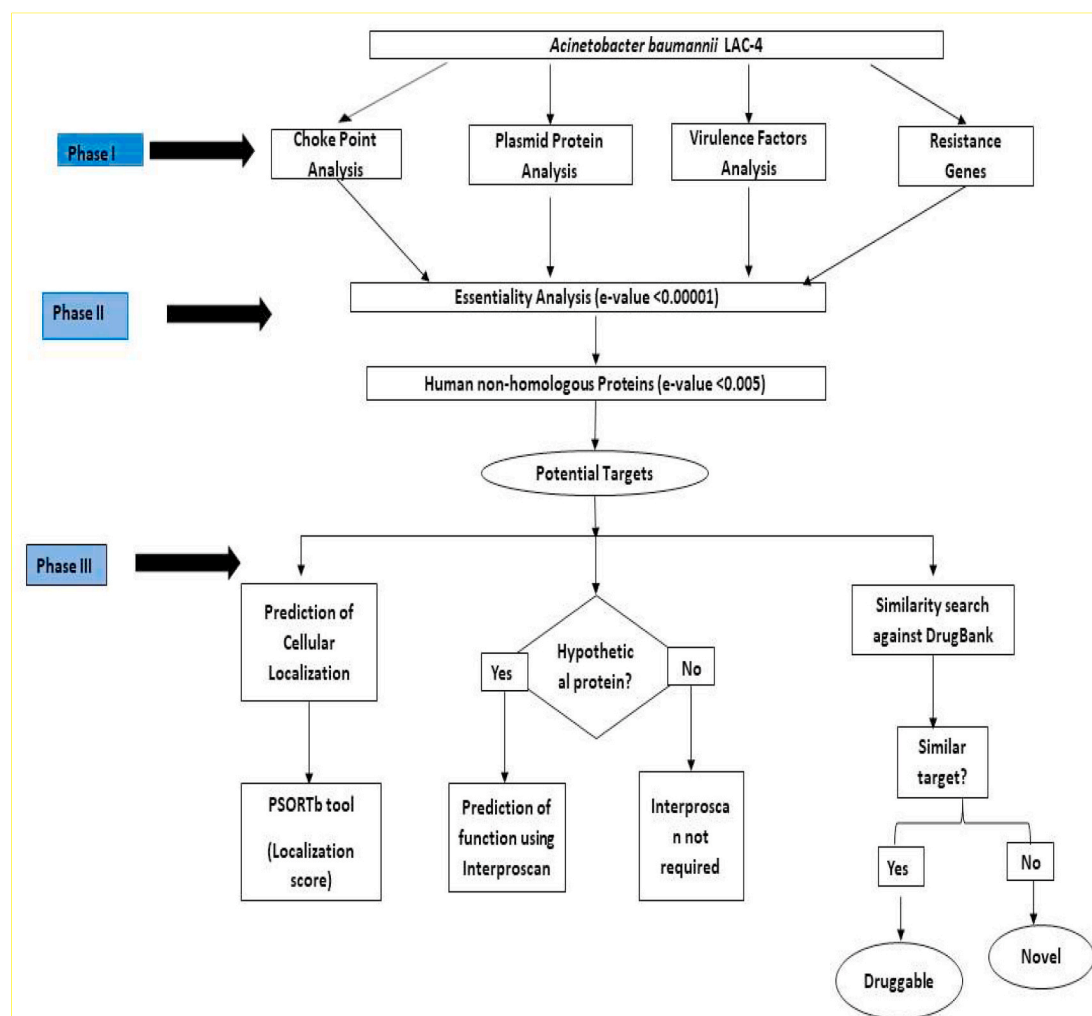


Fig. 1. Schematic flowchart of the study. Present study was carried out in three phases comprising of different analysis, in phase I unique proteins from different pathways were analyzed for choke points, also proteins from the plasmids and those responsible for virulence and resistance were shortlisted and combined for the analysis in phase II where they were analyzed for their essentiality in microorganism and homology with the human host and proteins. In phase III scrutinized proteins were evaluated for their qualitative characteristics like cellular location, function of the protein and druggability.

found in human. Chokepoint analysis was performed using simple statistics. Non-homologous proteins (enzymes) were sorted on the basis of confidence level, which was calculated as follows [16].

Confidence level (CL) = No. of pathways in which an enzyme is found as choke point/Total no. of pathways of the enzyme X100.

Proteins with a confidence score less than 50% were discarded and those with a score greater than 50% were selected for the protein-protein interaction studies.

2.1.3. Plasmid protein analysis

A. baumannii harbours two plasmids **pABLAC1** and **pABLAC2** that possess 17 protein-coding genes [17]. Plasmid encoding proteins are also a part of the resistome of the pathogen, and hence they can also be used as a drug target. Sequences of genes and proteins encoded by them were retrieved from the NCBI nucleotide database and further passed through subtractive channel of analysis.

2.1.4. Virulence factors analysis

Virulence Factor Database (VFDB) contains information about the virulence factors of 24 different genera of bacteria, including *Escherichia coli*, *Shigella*, *Salmonella*, *Chlamydia*, and *Mycoplasma* [18]. The whole list of virulence factors and genes encoding them were downloaded from the VFDB. The list of virulence proteins that are present in *A. baumannii* was retrieved from the KEGG/NCBI database. These proteins were further examined for their presence in human pathways manually, and proteins absent in the host and present in the pathogen alone were selected for the further subtractive channel of analysis.

2.1.5. Resistance genes and protein network analysis

Genes responsible for the drug resistance were obtained from the comprehensive antibiotic resistance database (CARD, <https://card.mcm.aster.ca/>) [19]. Sequences of proteins that are involved in the drug resistance were retrieved from the KEGG/NCBI database. All the proteins from the different steps of the phase I were clustered and subjected to the next phase of analysis.

2.2. Phase II: Subtractive channel of analysis

Shortlisted proteins were then analyzed for selectivity, specificity, and essentiality in phase II which is also known as subtractive channel of analysis as it only selects the proteins which are essential to the pathogen or non-homologous to host.

2.2.1. Essentiality analysis

BLASTp search was performed on the shortlisted proteins against Database of Essential Genes (DEG) in order to identify essential proteins among them [14,20]. Database of essential genes is a consortium of the gene sequences that are essential for the survival of the microorganism. Only those proteins which have E value $\leq 10^{-100}$, Bit score >100 were considered as significant hits.

2.2.2. Non-homology analysis

The non-homology analysis was performed to identify the proteins that are specific for the pathogen and are absent in the human host. This step is crucial in the drug target identification as it minimizes the risk of cross-reactivity of the drug with similar proteins present in the host system [10]. BLASTp search for all the selected proteins from phase I was performed against the non-redundant human database with E value < 0.005, identity <25%, Query length >30% [10].

2.3. Phase III: qualitative characterization of the shortlisted targets

2.3.1. Cellular localization

Defining the cellular location of the proteins is important as it provides information on whether the selected protein will behave as a drug or vaccine target. In the present study, PSORTb was used for the

prediction of the subcellular location of the target proteins. Three different modules support Vector Machines (SVM), open source transmembrane α -helix predictor (S-TMHMM), and SCL-BLAST of PSORTb were used for sorting proteins, utilizing information about known localization of 11692 proteins from bacteria (Gram-negative and positive) and archaea as the training set. PSORTb tool accepts protein sequences in FASTA format.

2.3.2. Druggability analysis

A target is considered druggable only if it has the ability to bind to drug candidates with high affinity. Targets selected from phase II were searched for homology against the drug targets present in the DrugBank 3.0 with an expect value of 10^{-5} [21] [22].

3. Results

Present study utilized subtractive genomics and comparative genomics approach for identification and characterization of potential drug targets in multi-drug resistant *A. baumannii*. The current target identification and characterization approach comprised of three phases of analysis. Phase I involved the mining of protein datasets through a chokepoint analysis, analysis of virulence factors, and resistance genes. Phase II consisted of a subtractive analysis of the mined protein datasets. By performing the sequence-level analysis, we filtered out proteins that were non-homologous to the human hosts. In the third phase predicted drug targets were characterized qualitatively for cellular location, and druggability.

3.1. Phase I

3.1.1. Pathway analysis

A comparison of metabolic pathways between host and pathogen reveals unique pathways that are specific for the pathogen. We analyzed 119 different metabolic pathways that are involved in different biosynthetic pathways like lipid, carbohydrate, amino acid, nucleotide, energy metabolism, amino acid, vitamin etc. Out of 119 pathways, only 36 pathways were identified as unique pathways (Table 1). Among these 36 pathways, non-homologous proteins were found in only 30 pathways, which corresponded to 185 proteins (enzymes), and some of these proteins were found in more than two pathways (see Table 2).

3.1.2. Chokepoint enzymes analysis

Out of 185 enzymes from different metabolic pathways, 90 enzymes were found as chokepoint enzymes with a confidence score of more than 50%. All the proteins with confidence score greater than 50% were analyzed for protein-protein interactions using STRING database. Almost all the proteins obtained after choke point analysis had 10 interactors. Only three of the choke point proteins namely ABLAC_23590, ABLAC_23260 and ABLAC_15870 had 9, 8 and 4 interactors respectively. Target proteins with maximum number of interactors are generally considered as metabolically active proteins.

3.1.3. Plasmid protein analysis

Sequences of 17 different plasmid encoding genes of *A. baumannii* namely Replicase C Aci3, RelE toxin of TA system, "Cro/C1 family of transcriptional regulator", 2'-aminoglycoside nucleotidyl transferase, protein MobA, "Protein MobC" pentapeptide repeat protein, hypothetical protein ABLAC_p100070, hypothetical protein ABLAC_p100090, hypothetical protein ABLAC_p100080, hypothetical protein ABLAC_p100040, Hypothetical protein with MerR superfamily HTH motif, hypothetical protein ABLAC_p100010, hypothetical protein ABLAC_p200070, hypothetical protein ABLAC_p200040 were retrieved from NCBI-nucleotide database.

3.1.4. Virulence factors analysis

Sixty-four virulence proteins were present in *A. baumannii*, and their

Table 1

Unique metabolic pathways.

Sr. No.	Name of Pathways	Pathway ID
1	Degradation of Aromatic compounds	aba101220
2	C5 Branched dibasic acid metabolism	aba100660
3	Methane Metabolism	aba100680
4	Secondary Bile acid Biosynthesis	aba100121
5	Phosphonate and Phosphinate metabolism	aba100440
6	Cyanoamino acid metabolism	aba100460
7	D-Alanine Metabolism	aba100473
8	Lipopolysaccharides Biosynthesis	aba100540
9	Peptidoglycan Biosynthesis	aba100550
10	Limonene and Pinene degradation	aba100903
11	Geraniol Degradation	aba100281
12	Biosynthesis of siderophore group non ribosomal peptides	aba101053
13	Non ribosomal peptide structures	aba101054
14	Carbapenem Biosynthesis	aba100332
15	Monobactam Biosynthesis	aba100261
16	Streptomycin Biosynthesis	aba100521
17	Novobiocin Biosynthesis	aba100401
18	Benzoate Degradation	aba100362
19	Aminobanazole Degradation	aba100627
20	Fluorobenzoate Degradation	aba100364
21	Chloroalkane and chloroalkene Degradation	aba100625
22	Chlorocyclohexane and chlorobenzene degradation	aba100361
23	Toluene Degradation	aba100623
24	Xylene Degradation	aba100622
25	Nitrotoluene Degradation	aba100633
26	Styrene Degradation	aba100643
27	Atrazene Degradation	aba100791
28	Caprolactam Degradation	aba100930
29	Naphthalene Degradation	aba100626
30	Phosphotransferase system	aba102060
31	Bacterial secretion system	aba103070
32	Two component System	aba102020
33	Quorum Sensing	aba102024
34	Beta Lactam Resistance	aba101501
35	Vancomycin Resistance	aba101502
36	Cationic Antimicrobial Peptide Resistance	aba101503

sequences were taken from the KEGG/NCBI database. Proteins that are specific for the pathogen were selected, and those homologous to host were discarded. List of different virulence factors selected include adeF, adeG, adeH, csuA, csuA/B, csuB, csuC, csuD, csuE, pgaA, pgaB, pgaC, pgaD, plcD, ACICU_0071, ACICU_0072, ACICU_0073, ACICU_0074, ACICU_0075, ACICU_0076, ACICU_0077, ACICU_0078, ACICU_0079, ACICU_0080, ACICU_0081, ACICU_0082, ACICU_0083, ACICU_0084, ACICU_0085, ACICU_0086, ACICU_0087, ACICU_0088, ACICU_0089, ACICU_0091, ACICU_0092, PGM, lpsB, lpxB, lpxC, lpxD, lpxL, barA, barB, basA, basB, basC, basD, basF, basG, basH, basI, basJ, bauA, bauB, bauC, bauD, bauE, bauF, entE, bfmR, bfmS, abaI, abaR and pbpG.

3.1.5. Resistance proteins analysis

Six proteins that cause resistance in the organism were selected for next phase of analysis (Phase II). All the proteins from different analyses were combined and selected for next phase of analysis.

3.2. Phase II: Subtractive analysis

3.2.1. Essentiality analysis

The combined proteins from phase 1 were screened against a database of essential genes with an expected value of 0.0001. Of 222 input proteins identified as best targets from phase I, only 68 proteins were observed to have threshold value less than 0.0001 and hence these 68 were considered as essential proteins, and others which don't have any hits were confirmed non-essential and omitted from the study.

3.2.2. Non-homology analysis

Non-homology of these 68 proteins was studied by similarity search against proteome of *H. sapiens* using BLASTp. Proteins with expected value less than 0.005 were considered as non-homologous and used for

Table 2

List of the druggable targets and the pathways they are involved and their function in bacterial cell.

Protein Enzyme code	Gene name	Property	Pathways Involved	Function
Acinetobacter baumannii LAC-4: ABLAC_01000	K07657 (phoB)	Druggable	Two component system	Resistance
Acinetobacter baumannii LAC-4: ABLAC_02740	K07659 (ompR)	Druggable	Two component system	Resistance
Acinetobacter baumannii LAC-4: ABLAC_31030	K07661 (rstA)	Druggable	Two component system	Resistance
Acinetobacter baumannii LAC-4: ABLAC_06010	K07664 (baeR)	Druggable	Two component system	Resistance
Acinetobacter baumannii LAC-4: ABLAC_12900	K07665 (cusR)	Druggable	Two component system, Antimicrobial Resistance Genes	Resistance
Acinetobacter baumannii LAC-4: ABLAC_13680	K07667 (kdpE)	Druggable	Two component system, Quorum Sensing	Resistance
Acinetobacter baumannii LAC-4: ABLAC_01630	K01921 (ddl)	Druggable	D-Alanine Metabolism, Peptidoglycan Biosynthesis, Vancomycin Resistance	Choke point
ACICU_00089	K00012 (UGDH)	Druggable	Amino Sugar And nucleotide sugar metabolism	Virulence factor
bfmR	bfmR	Druggable	A response regulator	Virulence factor
pbpG	pbpG	Druggable	Penicillin Binding Protein	Virulence factor

the further study whereas those with the expect value above 0.005 were homologous and excluded from the study. Among 68 proteins, 10 proteins were observed as close homologs and 58 were found to be non-homologous. 58 non-homologous proteins were regarded as potential drug targets while homologous proteins were excluded from the study.

3.3. Phase III: qualitative characterization of the selected drug targets

3.3.1. Cellular localization analysis

Classification of these 58 drug targets as probable drug or vaccine targets was done on the basis of their subcellular location which was predicted using PSORTb tool. PSORTb identified 32 proteins as cytoplasmic, 17 as cytoplasmic-membrane 2 as extracellular and 1 as periplasmic on the basis of their predicted localization score as demonstrated by the pie chart in Fig. 2. Cello2 go identified 33 proteins as cytoplasmic, 15 as inner membrane, 2 as extracellular, 4 as outer membrane and 4 as periplasmic proteins on the basis of their localization score.

3.3.2. Druggability analysis

BLASTp search against Drug Bank targets revealed that 18 proteins were similar with targets present in the drug bank and the remaining 40 drug targets were absent and therefore regarded as novel targets as shown in Fig. 3. Targets that shows hits against targets from Drug Bank were taken as druggable.

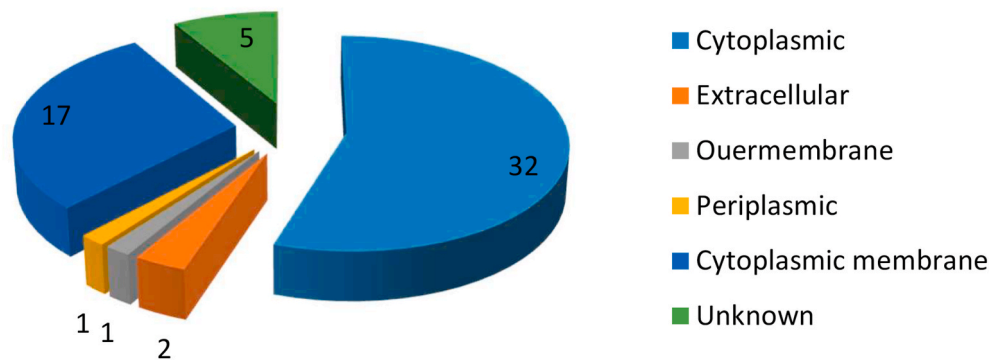


Fig. 2. Cellular location of potential drug targets.

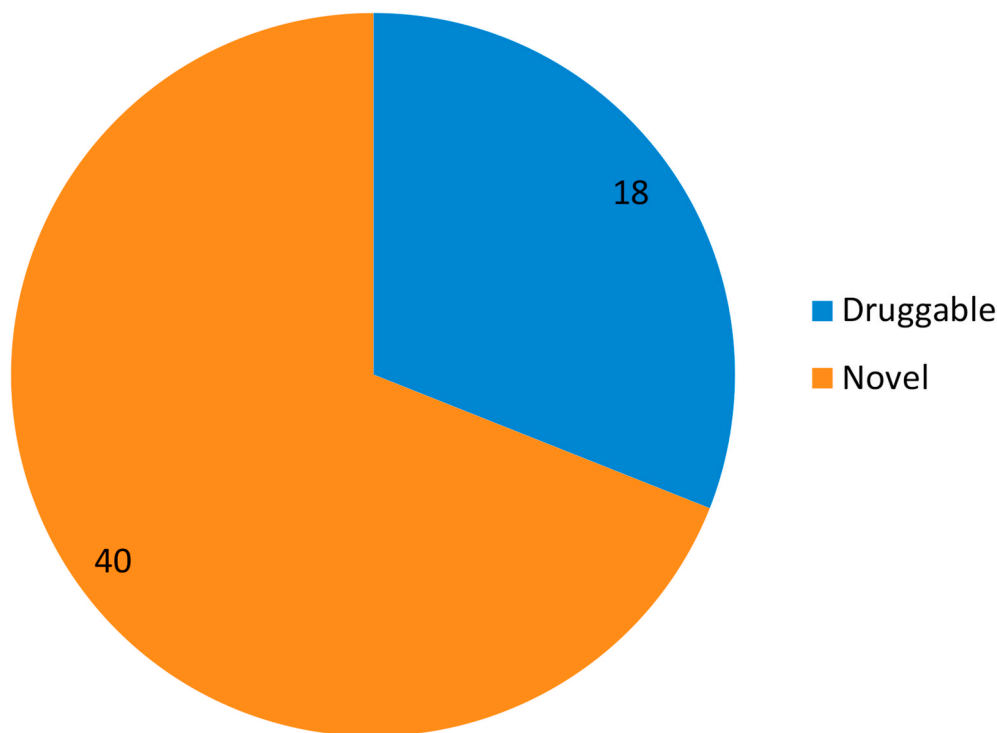


Fig. 3. Pie chart depicting the druggable property of potential drug targets.

4. Discussion

Multi-drug resistant *A. baumannii* has emerged as a therapeutic challenge. There is an urgent need to find effective chemotherapeutic agents to treat serious infections caused by this organism. As conventional drug discovery takes a long time, different approaches like virtual screening of herbal inhibitors, nanotherapeutics, and *in silico* approaches are being explored to find novel drug targets that may have potential to treat *A. baumannii* infections. Target identification is a crucial step as efficacy and selectivity of the target are deciding factors in the success or failure of the drug. Target identification could be achieved by using OMICS technologies or experimental methods like RNA interference and single-gene knockouts that in contrast to OMICS technology are very costly, and also tedious processes [23].

The approach of OMICS technologies has remodelled the idea of identification of drug targets in the drug discovery process [24]. Computational applications coupled with the availability of sequenced genomes make the research for new drug targets a novel and straightforward strategy in the drug discovery process. Finding putative drug targets by genomics applications primarily depends on those proteins

that are not found in the human host but present in the pathogenic organism as this will minimize the adverse effects on the human host [23]. Computational and subtractive genomics, along with metabolic pathway analysis, aids the identification of drug targets that are essential for the survival of pathogens and are absent in humans. Essential proteins that are important for the survival of pathogen is the priority in drug target identification process. Non-homologous targets were identified by subtracting the human genome from essential genes of the pathogen, which further prevents the cross-reactivity of drugs. Comparative genomics selects only those proteins which are highly conserved among different species. Applications of genomics are beneficial and have been used mainly in search of significant potential drug targets across a large number of organisms like *Mycobacterium tuberculosis* and *Pseudomonas aeruginosa* [25].

In this study, we combined two approaches: comparative and subtractive genomics for drug target identification. Comparative analysis of metabolic pathways in the present study identified 36 unique pathways. However, a study by Goyal et al. has identified 12 pathways as unique among the total 66 pathways they have studied in *Acinetobacter baumannii* [26]. This difference in targets identified in both studies might be

due to the difference in the approaches used in the study; they have used subtractive genomics alone, whereas we have used comparative as well as subtractive genomics for identification of unique pathways. Some of these unique pathways are studied extensively in bacteria like two-component systems, peptidoglycan synthesis, amino acid metabolism, and lipopolysaccharide synthesis [15]. Hence, selecting targets from these pathways might be applicable for the multiple bacterial species as these pathways are common in gram-negative bacteria. Anishetty et al., identified 6 unique pathways for drug target identification in *Mycobacterium tuberculosis* which included-proteins from lipid metabolism, carbohydrate metabolism, cofactor biosynthetic pathways, nucleotide metabolism and amino acid metabolism [15]. Singh et al. identified 16 unique pathways in *Acinetobacter bayli* [27]. Among 36 unique pathways identified in the present study, 185 non-homologous proteins were identified in 30 unique pathways. Chokepoint analysis of these proteins revealed 90 of them as choke point enzymes. Choke point analysis is a crucial step in drug target identification as proteins which have similar domains to that of human proteins may lead to side effects [28].

Possessing properties like being unique, non-homologous or involvement in metabolic pathways do not make a protein suitable for being selected as a drug target. Targeting proteins that are non-homologous to humans and are involved in multiple metabolic pathways may not always be beneficial as these proteins may have isoforms or may not be essential for the survival of pathogen. Therefore, pathogen proteins that fulfil the criteria of being unique and essential at the same time represent more attractive drug targets. Hence proteins collected from the different analysis of phase1 were combined and assessed for their essentiality in *A. baumannii*. Essentiality in itself is very precise step for the identification of drug targets as proteins that are not essential for the pathogen are of no use as a target. Therefore, among 222 proteins clustered after the phase 1 only 68 were found essential for *A. baumannii*. Another major factor in deciding target for drug discovery is non-homology, as targeting homologous proteins may lead to adverse effects on the humans. The non-homology analysis further narrowed the list of the predicted drug targets to 58. These two steps of the phase2 are focused on the selectivity and specificity. Selectivity was confirmed by the non-homology analysis, whereas specificity is addressed by essentiality analysis. A subtractive genome analysis method that we employed has been effectively applied in various organisms such as *Mycobacterium abscessus* [7], *Mycobacterium ulcerans* [23] and *Staphylococcus aureus* N315 [28].

Protein localization in the cell is a deciding factor for the determination of suitability and efficacy of protein as a drug target. Proteins present in the membrane are not suitable as drug targets as they are difficult to purify and assay. Cytoplasm based proteins are therefore suitable as a drug target. In this aspect, phase3 analysis was carried out to provide more precision by studying the location of the targets in bacterial cell, functionality of the selected proteins, and druggability characteristics of the target. Analysis based on the subcellular localization of the proteins may uncover the mechanism of disease or aid in the identification of novel drug targets. Protein synthesis in bacteria usually occurs in the cytoplasm. Proteins are localized in extracellular space, periplasm, cytoplasmic membrane, and outer membrane in Gram-negative bacteria. In contrast, in the case of Gram-positive bacteria, proteins are localized mostly in the cell wall, cytoplasmic membrane, and extracellular space. Cytoplasmic proteins can be used as putative drug targets, whereas membrane proteins can serve as vaccine targets.

We found 58 non-homologous proteins that were involved in different pathways, namely peptidoglycan synthesis, two-component system, lipopolysaccharide biosynthesis, D-alanine metabolism, histidine metabolism. A similar approach for the identification of drug targets in *Acinetobacter* was used by Uddin et al., and they identified 86 non-essential genes by metabolic pathway analysis and non-homology analysis [29]. Our study is more refined as we have included the virulence genes, plasmid proteins and resistance genes in our study along

with the metabolic pathway analysis. Additionally, we have included non-homology analysis as well as druggability, functionality, essentiality, and cellular localization, which have provided detailed information (like their location in the cell) about the drug targets. Some of the predicted drug targets are explained below.

D-alanyl-alanine synthetase A (ddl) is an enzyme that catalyzes the synthesis of D alanyl-D-alanine (D-ala-D-ala) an essential component of the cell wall [30]. D-ala-D-ala provides stability to the cell wall by crosslinking the peptide chain of peptidoglycan. Ligand binding abilities of the prokaryotic cell wall are controlled by D-alanine and is absent in eukaryotes. Therefore, inhibiting the enzyme ddlA can significantly weaken the integrity of the cell wall and leads to the osmotic lysis of the bacteria.

A previous study has reported ddl to be an exciting target for drug identification in *Mycobacterium tuberculosis* [31]. Known antibiotic, Vancomycin targets ddl, by making a balanced complex with the target protein and blocks the mechanism of cell wall synthesis, finally leading to lysis of cell division [32]. Penicillin-binding proteins (PBPs), which play a crucial role in peptidoglycan synthesis, are exploited as targets for β -lactams. Increased resistance for β -lactams in *A. baumannii* was associated with the alteration in PBPs.

Besides PBPs, other enzymes involved in peptidoglycan synthesis, which were identified as unique targets, included MurA, MurB, MurC, MurD, MurE and MurF. UDP-N-acetyl glucosamine 1-carboxyvinyltransferase (MurA) and UDP-N-acetylenolpyruvoylglucosamine reductase (MurB) catalyze the first and second step of peptidoglycan biosynthesis [33]. There is a disadvantage in using MurA as a target as MurA is encoded by two different genes *murA1* and *murA2*, which are less than 60% similar and encode for enzymes with different active sites [28]. Thus they can switch between themselves in the absence of either of them [34]. Similarly, MurB also has an isoform, MurB2 which can lead to failure of the drug. MurC and MurD which catalyze third and fourth reactions of peptidoglycan synthesis are encoded by a single gene and therefore, can be considered as a putative drug target. Many of the antimicrobial molecules, such as beta-lactam antibiotics and glycopeptides interfere with peptidoglycan synthesis (35).

We also identified proteins from the two-component system (TCS) as drug targets. TCS plays a vital role in sensing and adapting of *A. baumannii* to adverse environmental conditions [35]. In *A. baumannii* TCS were reported to contribute in the development of antibiotic resistance and virulence [36]. Exploring the molecular mechanism behind TCS and the regulatory proteins involved in pathways will reveal new drug targets for the *A. baumannii*. BfmRS, the response regulator of TCS plays a significant role in biofilm formation [37]. Biofilm formation plays an important role in the development of the resistance in microorganisms by forming a protective EPS layer which prevents the action of antibiotics on microorganisms. Biofilm formation by *A. baumannii* requires expression of pili, which helps in the attachment of the microorganism to the surface. The expression of pili in *A. baumannii* is regulated by *csu* operon, which in turn is under the control of BfmRS [37]. Knockout mutants of BfmRS in *A. baumannii* ATCC19606 were defective in biofilm formation [38]; hence proteins of BfmRS TCS can prove to be excellent targets for the treatment of infections caused by the *A. baumannii*. Thompson et al. and Milton et al. have reported the inhibitory effect of the 2-aminoimidazole compounds on functions of BfmRS on biofilm formation [39,40].

Enzymes lpxA, lpxC and kdsB, and kdsA from the LPS biosynthetic pathway were identified as drug targets. lpxA and lpxC are involved in the synthesis of LipidA, which is a part of LPS and is present in the conserved core oligosaccharide region. LipidA mediates the toxic effects of Gram-negative bacteria. KdsB is responsible for joining of KDO (sugar) to cytosine monophosphate which is present in the inner core of the LPS. These proteins are considered as drug targets because they are essential for the synthesis of lipidA which is responsible for the intact structure of the cell wall. However, a study by Bojkovic et al. reported that *A. baumannii* strains could tolerate the complete loss of

lipopolysaccharide, which raises the question on the essentiality of these proteins and hence on their probability of being a probable drug target [41].

The potential drug targets identified in this study are unique (BfmRS, kdsA, kdsB) and have the potential to be exploited for designing new antimicrobial agents against *A. baumannii*. Computer-aided drug design approach can further be used to identify hits and lead compounds against these targets. Computational and subtractive genomics coupled with metabolonomics guided the search for the identification of potential drug targets in *A. baumannii*. The present study shortlisted targets on the basis of their essentiality, non-homology to the human host, presence in drug bank, and subcellular location. Our study revealed potential drug targets that are key to the treatment of *A. baumannii* infections. The ubiquitous nature of the identified drug targets makes them permissible targets for drug discovery programs. The strategy utilized in the present study can be used for the identification of drug targets in other MDR human pathogens. Also, new drugs specific for the treatment of infections caused by *A. baumannii* can be discovered using these targets.

CCRediT authorship contribution statement

Harpreet Kaur: Concept or hypothesis of the paper, Methodology, Investigation, Formal analysis, Writing - original draft, Funding acquisition. **Manmohit Kalia:** Methodology, Investigation, Formal analysis, Writing - original draft, Funding acquisition. **Neelam Taneja:** Concept or hypothesis of the paper, Formal analysis, Writing - review & editing, Supervision.

Declaration of competing interest

Authors declare that no conflict of interest exists among them.

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